

Campus B5 5mm x215

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KOKUYO

测试题 1

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成绩 _____

一、填空题 (每小题 3 分, 共 15 分)

1. $\lim_{(x,y) \rightarrow (0,0)} \frac{\sqrt{x^2+y^2} - \ln(1+\sqrt{x^2+y^2})}{e^{x^2+y^2} - 1} = \underline{\frac{1}{2}}$

$$\lim_{t \rightarrow 0} \frac{t - \ln(1+t)}{e^{t^2} - 1} = \lim_{t \rightarrow 0} \frac{t - \ln(1+t)}{t^2} = \lim_{t \rightarrow 0} \frac{1 - \frac{1}{1+t}}{2t}$$

$$= \lim_{t \rightarrow 0} \frac{\frac{t}{1+t}}{2t} = \lim_{t \rightarrow 0} \frac{1}{2+2t}$$

2. xOz 面上的曲线 $C: \begin{cases} x^2 + 3z^2 = 9 \\ y = 0 \end{cases}$ 绕 z 轴旋转一周所生成的旋转曲面方程 $x^2 + y^2 + 3z^2 = 9$

3. 设 $f(x)$ 为连续函数, $F(t) = \int_1^t dy \int_y^t f(x) dx$, 求 $F'(2) = \underline{f(2)}$

4. 交换二次积分的积分次序 $I = \int_0^1 dx \int_{1-x}^1 f(x,y) dy + \int_1^e dx \int_{\ln x}^1 f(x,y) dy = \int_0^1 dy \int_{1-y}^e f(x,y) dx$

5. 已知 $\vec{m} = 5\vec{a} + \vec{b}$, $\vec{n} = \vec{a} - \vec{b}$, $|\vec{a}| = 2$, $|\vec{b}| = 1$, 向量 $\vec{a}$, $\vec{b}$ 的夹角 $\langle \vec{a}, \vec{b} \rangle = \frac{\pi}{4}$, 求以向量

$\vec{m}$, $\vec{n}$ 为边的三角形面积 $3\sqrt{2}$

二、计算题 (每小题 5 分, 共 30 分)

1. 设非零向量 $\vec{a}$, $\vec{b}$ 满足 $\vec{a} + 3\vec{b} \perp 7\vec{a} - 5\vec{b}$, $\vec{a} - 4\vec{b} \perp 7\vec{a} - 2\vec{b}$, 求向量 $\vec{a}$, $\vec{b}$ 的夹角 $\langle \vec{a}, \vec{b} \rangle$.

$$\begin{cases} 7\vec{a}^2 + 16\vec{a}\vec{b} - 15\vec{b}^2 = 0 & \textcircled{1} \\ 7\vec{a}^2 - 30\vec{a}\vec{b} + 8\vec{b}^2 = 0 & \textcircled{2} \end{cases}$$

$$\Rightarrow 46\vec{a}\vec{b} - 23\vec{b}^2 = 0$$

$$2\vec{a}\vec{b} = |\vec{b}|^2$$

$$\therefore \cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} = \frac{\frac{|\vec{b}|^2}{2}}{|\vec{a}| |\vec{b}|} = \frac{|\vec{b}|}{2|\vec{a}|} = \frac{1}{2}$$

$$\therefore \langle \vec{a}, \vec{b} \rangle = \frac{\pi}{3}$$

2. 求曲线 $\begin{cases} x^2 + y^2 + z^2 = 6 \\ z = x^2 + y^2 \end{cases}$ 在点 $(1, 1, 2)$ 处的切线方程与法平面方程.

$$\begin{cases} 2x + 2y \cdot \frac{dy}{dx} + 2z \cdot \frac{dz}{dx} = 0 \\ \frac{dz}{dx} = 2x + 2y \frac{dy}{dx} \end{cases}$$

$$\therefore \text{切线} \begin{cases} \frac{x-1}{1} = \frac{y-1}{-1} \\ z=2 \end{cases}$$

$$\Rightarrow \begin{cases} \frac{dy}{dx} = -1 \\ \frac{dz}{dx} = 0 \end{cases}_{(1,1,2)}$$

$$\text{法平面: } (x-1) - (y-1) = 0 \text{ 即 } x-y=0$$

3. 求在曲面 $z = \sqrt{-x^2 - y^2}$ 上与平面 $x + y - z + 3 = 0$ 平行的切平面的方程.

$$\text{由题得: } \vec{n} = (1, 1, -1)$$

$$\therefore (x + \frac{1}{2}) + (y + \frac{1}{2}) - (z - \frac{1}{2}) = 0$$

$$\text{令 } F(x, y, z) = x^2 + y^2 + z - 1$$

$$\text{即 } x + y - z + \frac{3}{2} = 0$$

$$\therefore \vec{r} = (F_x, F_y, F_z) = (2x, 2y, 1)$$

$$\therefore x = -\frac{1}{2}, y = -\frac{1}{2}, z = \frac{1}{2}$$

4. 求函数 $u = \ln(x + \sqrt{y^2 + z^2})$ 在点 $A(1, 0, 1)$ 处沿点 A 指向 $B(3, -2, 2)$ 方向的方向导数.

$$\begin{aligned} \vec{l} &= \vec{AB} = (2, -2, 1) \\ \vec{e} &= \left(\frac{2}{\sqrt{6}}, -\frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right) \\ u'_x|_{(1,0,1)} &= \frac{1}{2} \\ u'_y|_{(1,0,1)} &= 0 \\ u'_z|_{(1,0,1)} &= \frac{1}{2} \end{aligned}$$

5. 已知连续函数 $z = f(x, y)$ 在 $(0, 1)$ 处可微, 满足 $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 1}} \frac{f(x, y) - 2x + y - 2}{\sqrt{x^2 + (y-1)^2}} = 0$, 则求 $dz|_{(0,1)}$.

$$\begin{aligned} \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \frac{\Delta z - A\Delta x - B\Delta y}{\sqrt{(\Delta x)^2 + (\Delta y)^2}} &= \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \frac{f(\Delta x, 1 + \Delta y) - f(0, 1) - A\Delta x - B\Delta y}{\sqrt{(\Delta x)^2 + (\Delta y)^2}} \\ \text{又: } \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 1}} \frac{f(x, y) - 2x + y - 2}{\sqrt{x^2 + (y-1)^2}} &= 0 \end{aligned}$$

$i: \Delta x = x, \Delta y = y-1$
 $A=2, B=-1, f(0,1)=1$
 $dz|_{(0,1)} = 2dx - dy$

6. 设 $y = f(x, t)$, 而 $t = t(x, y)$ 是由方程 $F(x, y, t) = 0$ 所确定的函数, 其中 f, F 都具有

一阶连续偏导数, 求 $\frac{dy}{dx}$.

证法一 由方程组 $\begin{cases} y = f(x, t) \\ F(x, y, t) = 0 \end{cases}$ 可确定两个一元隐函数 $y = y(x), t = t(x)$. 分别

在两个方程两端对 x 求导可得

$$\begin{cases} \frac{dy}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial t} \cdot \frac{dt}{dx} \\ \frac{\partial F}{\partial x} + \frac{\partial F}{\partial y} \cdot \frac{dy}{dx} + \frac{\partial F}{\partial t} \cdot \frac{dt}{dx} = 0 \end{cases}$$

移项得

$$\begin{cases} \frac{dy}{dx} - \frac{\partial f}{\partial t} \cdot \frac{dt}{dx} = \frac{\partial f}{\partial x} \\ \frac{\partial F}{\partial y} \cdot \frac{dy}{dx} + \frac{\partial F}{\partial t} \cdot \frac{dt}{dx} = -\frac{\partial F}{\partial x} \end{cases}$$

$$\text{当 } D = \begin{vmatrix} 1 & -\frac{\partial f}{\partial t} \\ \frac{\partial F}{\partial y} & \frac{\partial F}{\partial t} \end{vmatrix} = \frac{\partial F}{\partial t} + \frac{\partial f}{\partial t} \cdot \frac{\partial F}{\partial y} \neq 0 \text{ 时, 解方程组得}$$

$$\frac{dy}{dx} = \frac{1}{D} \cdot \begin{vmatrix} \frac{\partial f}{\partial x} & -\frac{\partial f}{\partial t} \\ -\frac{\partial F}{\partial x} & \frac{\partial F}{\partial t} \end{vmatrix} = \frac{\frac{\partial f}{\partial x} \cdot \frac{\partial F}{\partial t} - \frac{\partial f}{\partial t} \cdot \frac{\partial F}{\partial x}}{\frac{\partial F}{\partial t} + \frac{\partial f}{\partial t} \cdot \frac{\partial F}{\partial y}}$$

三、计算下列各题 (每小题 7 分, 共 28 分)

1. 已知函数 $f(x, y)$ 满足 $f''_{xy}(x, y) = 2(y+1)e^x$, $f'_x(x, 0) = (x+1)e^x$, $f(0, y) = y^2 + 2y$,

求 $f(x, y)$ 的极值.

由题得: $f'_x(x, y) = (y^2 + 2y + x + 1)e^x$

$$f(x, y) = (y^2 + 2y + x)e^x$$

$$\begin{cases} (y^2 + 2y + x + 1)e^x = 0 \\ 2e^xy + 2e^x = 0 \end{cases} \Rightarrow \begin{cases} y^2 + 2y + x + 1 = 0 \\ 2y + 2 = 0 \end{cases}$$

$$\therefore \begin{cases} x_0 = 0 \\ y_0 = -1 \end{cases}$$

$$A = f''_{xx}(x_0, y_0) = (y^2 + 2y + x + 2)e^x|_{(0, -1)} = 1$$

$$B = f''_{xy}(x_0, y_0) = 2e^xy + 2e^x|_{(0, -1)} = -2 + 2 = 0$$

$$C = f''_{yy}(x_0, y_0) = 2e^x|_{(0, -1)} = 2$$

$\therefore AC - B^2 = 2 > 0$, 存在极小值

$$f(0, -1) = -1$$

② 内部极值

$$x^2 + y^2 - 18 = 0$$

$$\begin{cases} 4 - 2x = 0 \\ -4 - 2y = 0 \end{cases} \Rightarrow \begin{cases} x_0' = 2 \\ y_0' = -2 \end{cases}$$

2. 求 $f(x, y) = 4x - 4y - x^2 - y^2$ 在区域 $D: x^2 + y^2 \leq 18$ 上的最大值和最小值。

① 边界极值 设 $L(x, y, \lambda) = 4x - 4y - x^2 - y^2 + \lambda(x^2 + y^2 - 18)$

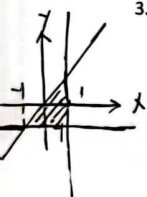
$$f(2, -2) = 8 + 8 - 4 - 4 = 8$$

$$\begin{cases} L_x = 4 - 2x + 2\lambda x = 0 \\ L_y = -4 - 2y + 2\lambda y = 0 \\ L_\lambda = x^2 + y^2 - 18 = 0 \end{cases}$$

$$\therefore \begin{cases} x_0 = 2 \\ y_0 = -2 \end{cases} \text{ 或 } \begin{cases} x_0' = -2 \\ y_0' = 2 \end{cases}$$

$$\begin{cases} \text{Max} = 8 \\ \text{Min} = 6 \end{cases}$$

$$f(2, 2) = f(-2, -2) = 12 + 12 - 9 - 9 = 6$$



3. $I = \iint_D y[1 + xe^{\frac{1}{2}(x^2+y^2)}] d\sigma$, 其中 D 由直线 $y=x$, $y=-1$ 及 $x=1$ 所围成的区域。

$$I = \int_{-1}^1 dx \int_{-1}^x [y + xy e^{\frac{1}{2}(x^2+y^2)}] dy$$

$$= \frac{x^3}{6} + \frac{e^{x^2}}{2} - \frac{x}{2} - e^{\frac{1}{2}(x^2+1)}$$

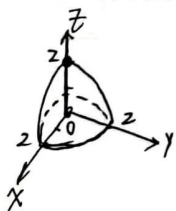
$$= \int_{-1}^1 \left[\frac{y^2}{2} + x e^{\frac{1}{2}(x^2+y^2)} \right]_{-1}^x dx$$

$$= \frac{1}{6} + \frac{e}{2} - \frac{1}{2} - e + \frac{1}{6} - \frac{e}{2} - \frac{1}{2} + e$$

$$= \int_{-1}^1 \left[\frac{x^2}{2} + x e^{x^2} - \frac{1}{2} - x e^{\frac{1}{2}(x^2+1)} \right] dx$$

$$= -\frac{2}{3}$$

4. 计算 $I = \iiint_{\Omega} e^{-(z-2)^2} dv$, 其中 Ω 由曲面 $z = 2 - \frac{1}{2}(x^2 + y^2)$ 及 $z=0$ 围成的闭区域。



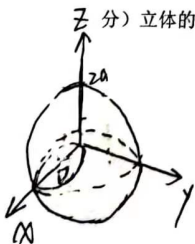
$$I = \int_0^2 dz \iint_{D_z} e^{-(z-2)^2} dx dy$$

$$= \int_0^2 [\pi (4-2z) e^{-(z-2)^2}] dz$$

$$\stackrel{t=z-2}{=} \int_{-2}^0 [\pi (-2t) e^{-t^2}] dt = \pi e^{-t^2} \Big|_{-2}^0 = \pi - \frac{\pi}{e^4}$$

四、应用题 (每小题 7 分, 共 14 分)

1. 求球体 $x^2 + y^2 + z^2 \leq 4a^2$ 被圆柱面 $x^2 + y^2 = 2ax$ ($a > 0$) 所截得的 (含在圆柱面内的部分) 立体的体积。(7 分)



$$V = 4V_0 = 4 \int_0^{\frac{\pi}{2}} \int_0^{2a \cos \theta} \sqrt{4a^2 - x^2 - y^2} dx dy$$

$$\begin{aligned} &= -\frac{32}{3} a^3 \left[\int_0^{\frac{\pi}{2}} \sin \theta d\theta + \int_0^{\frac{\pi}{2}} \cos^3 \theta d\theta \right] \\ &= -\frac{32}{3} a^3 \left(\frac{\pi}{2} - \frac{2}{3} \right) \end{aligned}$$

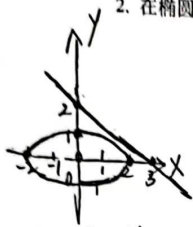
$$\therefore D = \begin{cases} 0 \leq \theta \leq \frac{\pi}{2} \\ 0 \leq \rho \leq 2a \cos \theta \end{cases}$$

$$\therefore \text{原式} = 4 \int_0^{\frac{\pi}{2}} d\theta \int_0^{2a \cos \theta} \sqrt{4a^2 - \rho^2} \rho d\rho$$

$$= -\frac{4}{3} \int_0^{\frac{\pi}{2}} \left[(4a^2 - \rho^2)^{\frac{3}{2}} \right]_0^{2a \cos \theta} d\theta = -\frac{32}{3} a^3 \int_0^{\frac{\pi}{2}} (\sin^3 \theta - 1) d\theta$$

$$x^2 + y^2 = 1$$

2. 在椭圆 $x^2 + 4y^2 = 4$ 上求一点，使其到直线 $2x + 3y - 6 = 0$ 的最短距离。



$$\text{设 } L(x, y) = \frac{1}{13} (2x + 3y - 6)^2 + \lambda (x^2 + 4y^2 - 4)$$

$$\begin{cases} \frac{1}{13} \times 4 (2x + 3y - 6) + 2\lambda x = 0 \\ \frac{1}{13} \times 6 (2x + 3y - 6) + 8\lambda y = 0 \\ x^2 + 4y^2 - 4 = 0 \end{cases}$$

$$\therefore d_{\min} = \frac{\sqrt{13}}{13}$$

$$d = \frac{|2x + 3y - 6|}{\sqrt{4 + 9}}$$

五、证明题 (共 13 分)

$$= \frac{|2x + 3y - 6|}{\sqrt{13}}$$

1. 设 $z = f(x^2 - y^2, 2xy)$ ，其中 $f(u, v)$ 具有二阶连续偏导数，且有 $\frac{\partial^2 f}{\partial u^2} + \frac{\partial^2 f}{\partial v^2} = 0$ 证明：

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0 \quad (6 \text{ 分})$$

$$\frac{\partial z}{\partial y} = -2y \frac{\partial f}{\partial u} + 2x \frac{\partial f}{\partial v}$$

$$\text{令 } u = x^2 - y^2, v = 2xy$$

$$\frac{\partial^2 z}{\partial y^2} = -2 \frac{\partial f}{\partial u} + 4y^2 \frac{\partial^2 f}{\partial u^2} + 4x^2 \frac{\partial^2 f}{\partial v^2}$$

$$\therefore \frac{\partial z}{\partial x} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial x}$$

$$= 2x \frac{\partial f}{\partial u} + 2y \frac{\partial f}{\partial v}$$

$$\therefore \frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = (4x^2 + 4y^2) \left(\frac{\partial^2 f}{\partial u^2} + \frac{\partial^2 f}{\partial v^2} \right)$$

$$\frac{\partial^2 z}{\partial x^2} = 2 \frac{\partial f}{\partial u} + 4x^2 \frac{\partial^2 f}{\partial u^2} + 4y^2 \frac{\partial^2 f}{\partial v^2}$$

$$= 0$$

2. 证明： $\iint_D f(x+y) dx dy = \int_{-1}^1 f(u) du$ ，其中闭区域 $D = \{(x, y) \mid |x| + |y| \leq 1\}$. (7 分)

$$\text{令 } u = x + y, v = x - y$$

$$\therefore x = \frac{u+v}{2}, y = \frac{u-v}{2}$$

$$D = \{(u, v) \mid -1 \leq u \leq 1, -1 \leq v \leq 1\}$$

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{vmatrix} = -\frac{1}{2}$$

$$\iint_D f(x+y) dx dy = \iint_{D'} f(u) |J| du dv$$

$$= \int_{-1}^1 f(u) du \int_{-1}^1 \frac{1}{2} dv = \int_{-1}^1 f(u) du$$

